

Evaluating Privacy-Utility Tradeoffs in Machine Learning

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1. Problem Description

Modern machine learning systems are increasingly trained on large-scale datasets containing sensitive user information, including medical records, financial data, and personal communications. While centralized training typically yields higher-performing models, it simultaneously exposes private data to a range of threats. The core challenge of Privacy-Preserving Machine Learning (PPML) is to achieve models of comparable utility while providing rigorous, quantifiable privacy guarantees. Rather than studying a single technique in isolation, this project proposes to benchmark three leading PPML approaches on a shared classification task, directly measuring their privacy-utility tradeoffs under controlled conditions.

System

The target system is a centralized or distributed ML training pipeline. To address these issues, a range of privacy-preserving machine learning techniques have been developed, including **differential privacy** (DP), which injects noise to provide formal privacy guarantees, **federated learning** (FL), which decentralizes training by keeping data on local devices, improving data locality and reducing direct exposure, and **homomorphic encryption** (HE), which allows computation directly on encrypted data.

Assets

The primary assets in this project are the training data and model parameters. The training data consists of personal records such as images or tabular features that are tied to sensitive attributes like health conditions, financial status, or personal behavior. In addition, the trained model parameters constitute a critical asset, as they may implicitly encode statistical patterns from the underlying data and can be exploited by adversaries through techniques such as membership inference or model inversion attacks.

Attackers

The threat model in this project contains multiple types of adversaries with varying capabilities and access levels. An honest but curious model owner adheres to the prescribed training protocol but may attempt to analyze intermediate outputs, such as gradients or activations, to infer sensitive information about the underlying training data. An external adversary interacts with the deployed model via query access, exploiting it to conduct attacks such as membership inference or model inversion to extract private information about individuals in the training dataset. In the federated learning setting, a malicious participant represents a compromised or adversarial client that deviates from the protocol by manipulating local updates, potentially aiming to reconstruct other participants' data or to introduce poisoned contributions that degrade the integrity of the global model. Collectively, these adversarial models reflect realistic and widely studied threat scenarios in privacy-preserving machine learning systems.

Vulnerabilities

The system has several key vulnerabilities that can lead to privacy leakage. One major issue is gradient leakage, where raw gradients from standard training can be used to reconstruct original training data. Another concern is memorization, as overparameterized models may unintentionally store and reproduce specific training samples. In federated learning, aggregation exposure can also occur, where even combined updates may reveal information about individual participants or their local data. These vulnerabilities demonstrate how sensitive information can be exposed at different stages of the machine learning process.

Threats

The primary threats to the system include **membership inference attacks** (MIA) and **model inversion attacks** (MIV). In a membership inference attack, an adversary attempts to determine whether a specific data record was included in the training set, thereby compromising individual privacy. In a model inversion attack, the adversary seeks to reconstruct representative input data from model outputs, potentially recovering sensitive information about the original training data. These threats highlight how trained models can unintentionally expose private information even without direct access to the underlying dataset.

Risks

The likelihood of MIA is high for models deployed via public APIs, as demonstrated by extensive empirical studies. MIV is moderately likely for face recognition and medical imaging models. Consequences range from regulatory violations (GDPR, HIPAA) and reputational damage to direct harm for individuals whose health or financial data is exposed. The primary risk of PPML techniques themselves is reduced model utility.

2. Proposed Approach

This project falls within the broad category of benchmarking and comparative evaluation of PPML techniques. We systematically evaluate three established privacy-preserving paradigms (DP, FL, and HE) on a shared image classification task, measuring both privacy guarantees and downstream utility.

Expected Benefits

Ultimately, this work provides a unified, reproducible benchmark that directly answers "which technique offers the best privacy-utility tradeoff?" under equivalent conditions. Through this evaluation, it also surfaces critical regime-specific insights, demonstrating that DP may best suit centralized settings, FL is preferable when data must remain on-device, and HE excels in inference-only scenarios.

Known Drawbacks

Each paradigm, however, presents distinct limitations. DP-SGD degrades accuracy substantially at tight privacy budgets, especially on complex tasks. Meanwhile, FL provides empirical rather than formal privacy guarantees and remains vulnerable to gradient inversion without additional DP. Furthermore, HE incurs extreme computational overhead, often being orders of magnitude slower, which effectively limits its application to small models or inference-only pipelines.

3. Final Product Description

Our final product will include a final report, presentation slides, source code, and experimental results comparing privacy-preserving ML approaches such as differential privacy, federated learning, and homomorphic encryption. The report will summarize related work, explain our experimental setup, compare utility and privacy-related metrics, and discuss practical limitations. The code will include a baseline model, DP experiments, FL simulation, and a simplified encrypted inference prototype or runtime comparison.

Milestone Plan

Week	Milestone
Weeks 3–4	Read core papers and finalize the dataset, baseline model, evaluation metrics, and the specific privacy-preserving methods to test.
Weeks 5–6	Run the main experiments: train the non-private baseline, test DP-SGD and FL simulation, compare with the baseline, and adjust settings as needed.
Weeks 7–8	Complete the simplified HE/encrypted inference prototype or runtime comparison. Summarize results into tables/figures and analyze tradeoffs.
Week 9	Write the draft report with introduction, related work, approach, preliminary results, evaluation, and limitations.
Final Week	Revise the report based on feedback, prepare slides, and include a brief demo, screenshot, or result visualization.

Responsibilities

Workstream	Lead Responsibility
DP	Lead DP-SGD experiments and baseline comparison.
FL	Lead FL simulation, client data split, and FedAvg comparison.
HE / Encrypted Inference	Lead simplified encrypted inference prototype or runtime comparison.

All members will contribute to writing, revision, result interpretation, and the final presentation.

4. References

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